GRADUATION PROJECT 2026

Nadel

Fully Autonomous Service Waiter

An Autonomous Restaurant Service Robot for Food Delivery, Navigation, and Human-Robot Interaction

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THE PROBLEM

The hospitality industry is facing an unprecedented operational crisis, directly impacting customer satisfaction and bottom-line revenue.

Long Waiting Times

Customers experience severe delays during peak hours, leading to frustration and lost retention.

Staff Shortages

A global deficit in hospitality workers is forcing restaurants to operate below optimal capacity.

Order Errors

Human error during high-stress service periods leads to wrong deliveries and food waste.

High Labor Costs

Rising wages and turnover rates drastically reduce the profit margins of restaurant owners.

COVID-19 Pandemic

The pandemic reshaped restaurant operations, increasing demand for safer, contactless, and more resilient service models.

COVID-19 exposed how fragile human-dependent restaurant operations can be, especially under health restrictions and labor shortages.

Contactless Service

Demand increased for QR menus, self-ordering, and minimal face-to-face interaction.

Staff Shortages

Lockdowns and health risks forced restaurants to operate with fewer workers.

Hygiene & Safety

Restaurants had to reduce close contact, shared surfaces, and repeated table visits.

Operational Disruption

Restaurants faced pressure to keep service quality while adapting to new restrictions.

OUR SOLUTION: NADEL

A Seamless Bridge from Kitchen to Customer:

- **Autonomous Navigation:** Safe movement in crowded spaces.
- **Food Delivery:** Secure transport from kitchen to precise table locations.
- **Voice Interaction:** Natural language processing for customer engagement.
- **Call/Pay Requests:** Integrated table modules for instant service.
- **Smart Workflow Integration:** Syncs perfectly with restaurant POS systems.

SYSTEM OVERVIEW

A fully integrated ecosystem communicating seamlessly via ROS2 and web protocols.

Customer
Interaction & Requests

⇄

ESP32 Table
Call / Pay Buttons

⇄

Nadel Robot
ROS2 Core & Sensors

⇄

Kitchen
Order Dispatch

Past HARDWARE Problems

- **NVIDIA Jetson Orin Nano:** The primary compute brain, handling intensive AI, SLAM, and computer vision tasks.
- **RPLIDAR A1M8:** Provides 360-degree 2D laser scanning for mapping and localization.
- **Intel RealSense D435i:** Depth camera used for 3D obstacle avoidance and perception.
- **ESP32 Microcontrollers:** Handles low-level motor controls and table module communications.
- **Ultrasonic Sensors & Encoders:** Provide short-range collision prevention and precise odometry data.
- **Display & Motors:** Interactive touchscreen face and high-torque DC motors for mobility.

HARDWARE ARCHITECTURE

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SOFTWARE ARCHITECTURE

Application Layer	Web UI / API
Voice AI Layer	Whisper + Gemma LLM
Restaurant Workflow Layer	Task Manager / State Machine
Navigation Layer	Nav2 / Path Planning
Perception Layer	SLAM Toolbox / Sensor Fusion
ROS2 Layer	ROS2 Humble Middleware
Hardware Abstraction	Serial / Drivers / micro-ROS

BUSINESS MODEL & VIABILITY

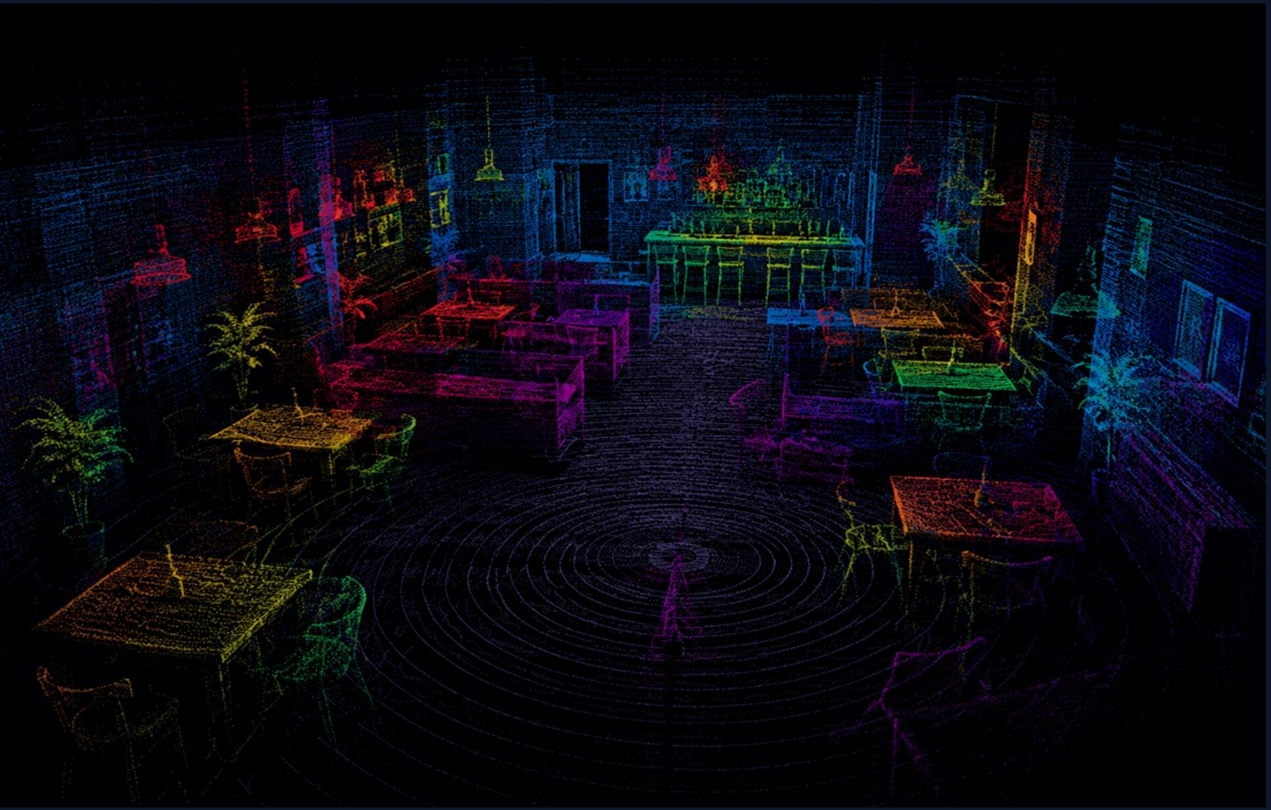
Target Market

- **Restaurants & Fast Casual:** Reducing overhead and wait times.
- **Cafes & Coffee Shops:** Automating repetitive table service.
- **Hotels:** Room service and lobby assistance.
- **Event Spaces:** Roaming beverage and appetizer delivery.

Revenue Streams

- **Direct Robot Sales:** Upfront hardware purchase.
- **RaaS (Robot as a Service):** Monthly leasing model.
- **Maintenance & Support:** Annual service contracts.

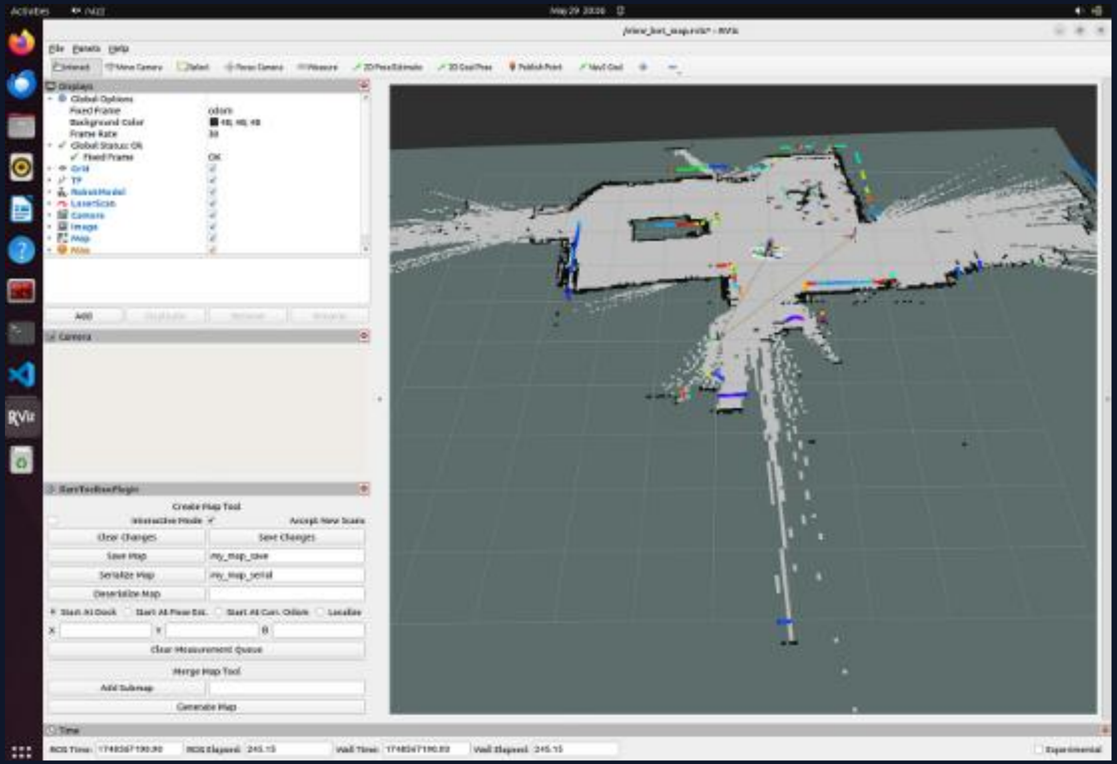
MAPPING & LOCALIZATION



To deliver food accurately, Nadel must know its exact coordinates on the restaurant floor down to the centimeter.

- **SLAM Toolbox:** Generates highly accurate 2D occupancy grids using LiDAR data during the initial mapping phase.
- **Occupancy Grid:** A digital blueprint defining free space, static obstacles, and restricted zones.
- **AMCL (Adaptive Monte Carlo Localization):** Probabilistically tracks the robot's pose against the known map using particle filters.
- **Odometry Fusion:** Combines wheel encoder data with LiDAR scans for smooth pose estimation.

SLAM TOOLBOX WORKFLOW



Initial environment representation is generated using the ROS2 SLAM Toolbox, operating in synchronous mapping mode.

- **Pose Graph Optimization:** Continuously refines the robot's trajectory by minimizing errors between sequential laser scans.
- **Loop Closure:** Algorithms detect previously visited areas to correct accumulated odometry drift.
- **Occupancy Grid (/map):** 2D representation where each pixel (0.05m resolution) is assigned a probability of being a static obstacle, free space, or unknown.

SENSOR FUSION & STATE ESTIMATION

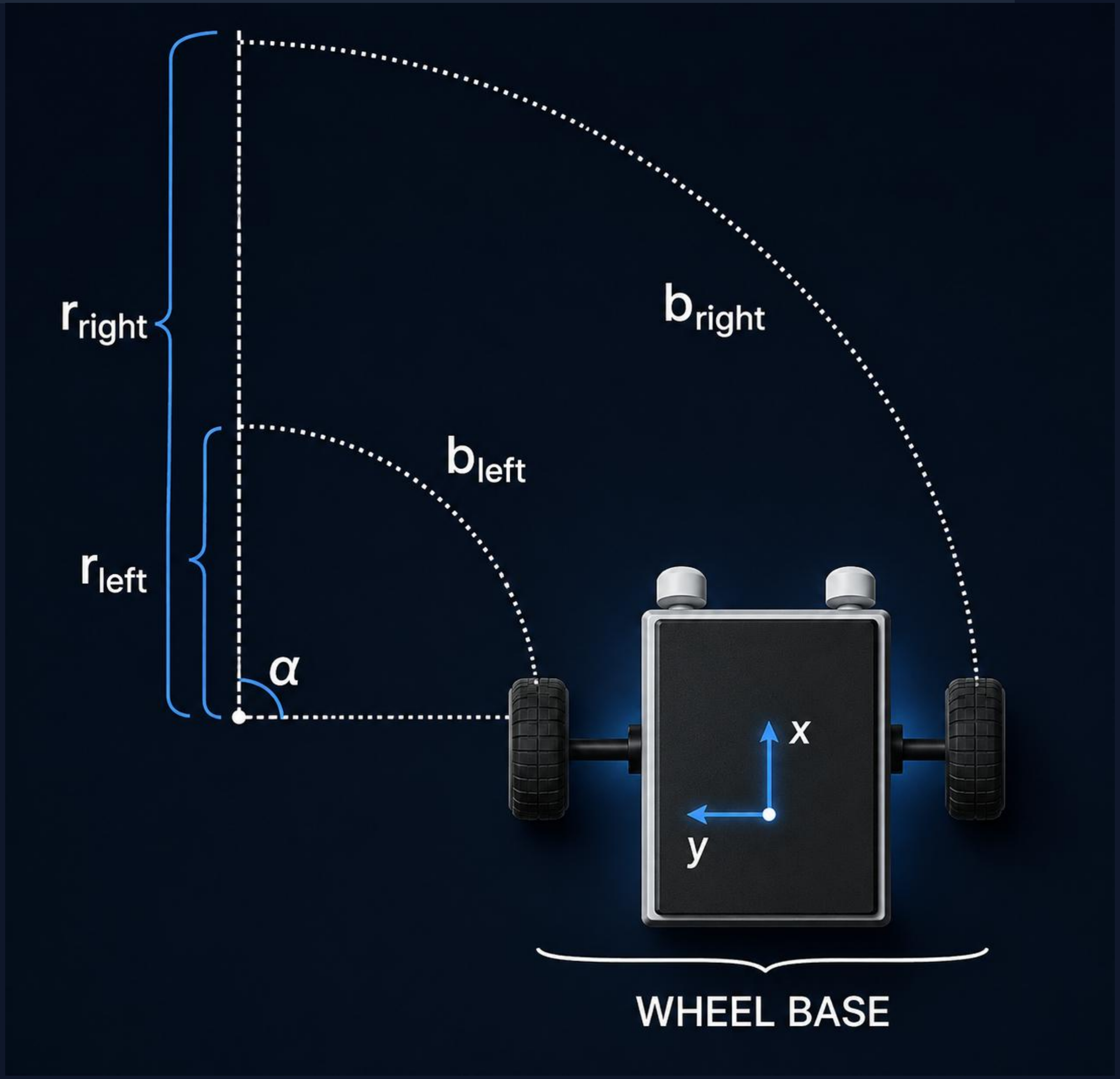
A multi-stage filtering approach ensures robust global positioning even during wheel slip.

Extended Kalman Filter (EKF):
Merges wheel encoder ticks with IMU data via robot_localization to output smooth odom → base_link transforms.

AMCL:
Adaptive Monte Carlo Localization uses a particle filter to match LiDAR scans against the map, correcting global position (map → odom).

EKF State Vector

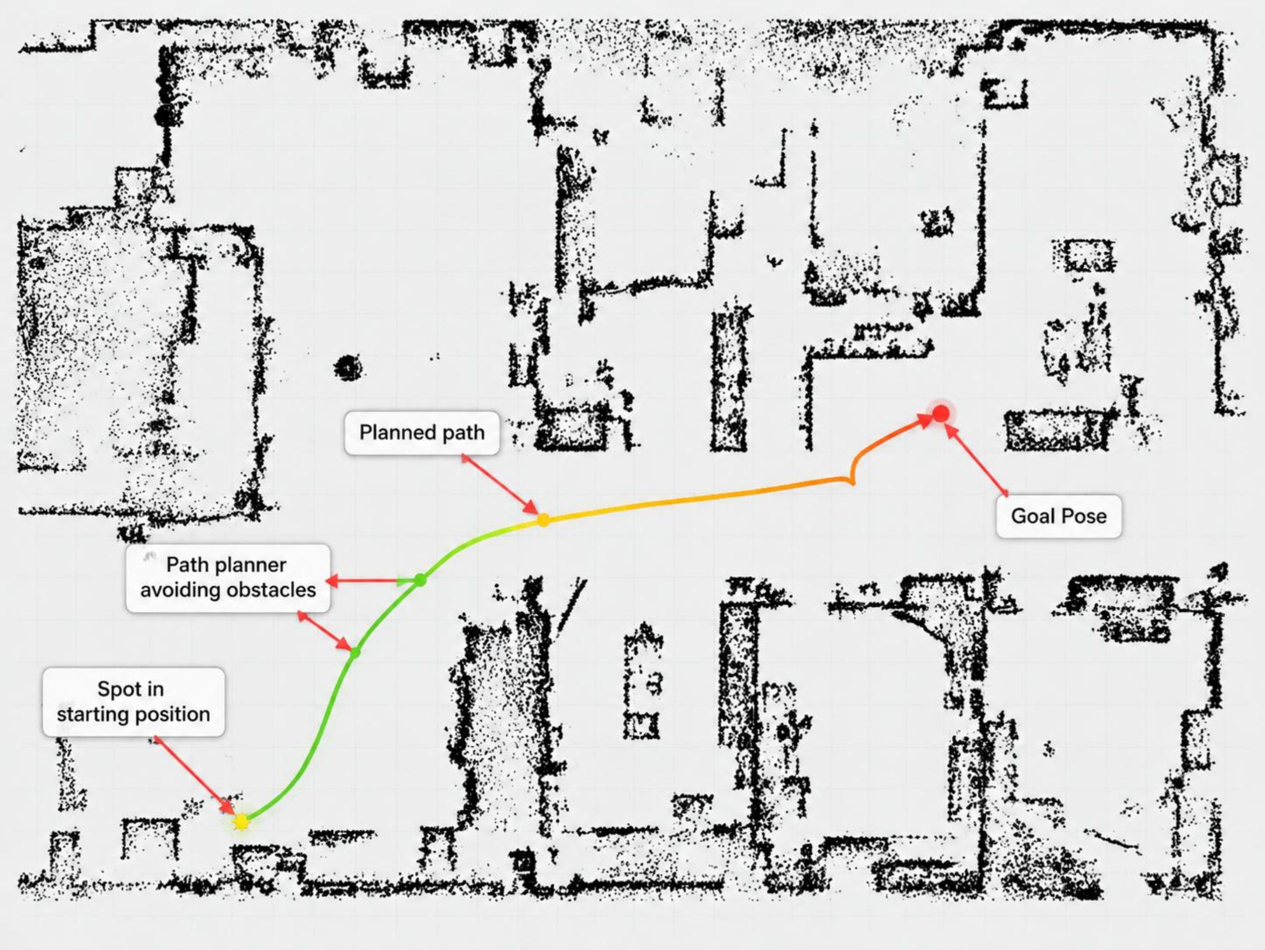
$$x_t = x , y , \theta , v_x , v_y , \omega^T$$



NAVIGATION SYSTEM

Nadel utilizes an advanced navigation stack designed for dynamic, unpredictable environments like busy restaurants.

- **ROS2 Humble:** The robust middleware enabling asynchronous node communication.
- **Nav2 Framework:** Handles the core logic of getting from point A to point B safely.
- **Path Planning:** Global planners map the shortest route, while local planners adjust dynamically in real-time.
- **Dynamic Obstacle Avoidance:** Instantly recalculates trajectories when patrons or chairs block the path.



PERCEPTION SYSTEM

Nadel sees the world through a multi-modal sensor array, ensuring no obstacle goes unnoticed.

- **2D LiDAR:** Primary sensor for long-range, 360-degree environmental profiling and map matching.
- **Ultrasonic Sensors:** Acts as a fail-safe proximity ring for immediate emergency stops.
- **Sensor Fusion:** ROS2 nodes aggregate data from all sources into a unified costmap for the navigation stack.

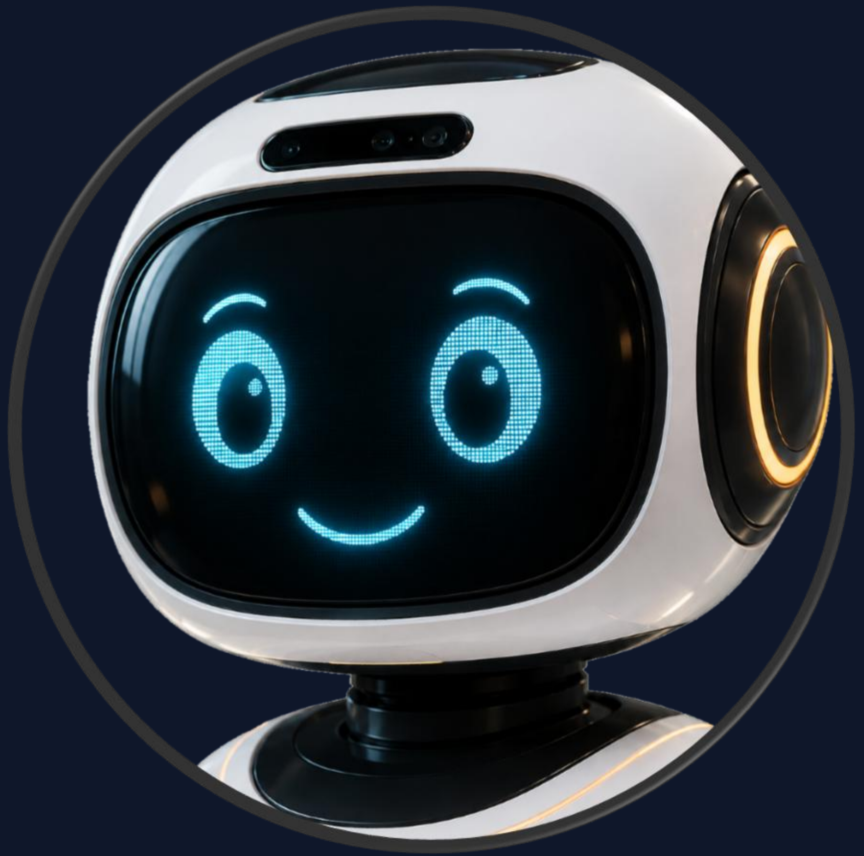
HUMAN-ROBOT INTERACTION

Bridging the Digital and Human

Face UI & Touch Screen: Nadel displays animated, expressive digital eyes to convey intent and friendliness, accompanied by a touch interface.

comfort: facial reactions to make the customers more comfortable dealing with Nadel

Interactive face: the face reacts to user order changing his facial expressions depending on the context.



User Interface & Interaction

Bridging the Digital and Human

Bilingual Interface: Fully supports both Arabic and English interfaces to cater to local and international patrons.

Customer Experience Focus: Designed to feel like a helpful assistant rather than a soulless machine, enhancing the dining atmosphere.

Voice: .

VOICE AI PIPELINE

An on-board, low-latency conversational AI pipeline.

1. Speech Input

Customer speaks request

2. STT Engine

Whisper transcribes audio

3. LLM Processing

Gemma understands intent

4. Generation

AI crafts response contextually

5. TTS Engine

Text converted to natural speech output

SMART RESTAURANT WORKFLOW

Order Placed

Via User Interface

Kitchen Prep

Chefs complete order

Dispatch

Robot assigned to table

Navigation

Autonomous routing

Delivery

Customer notified

Payment

Assistance provided

DESIGN CHALLENGE: IDENTIFYING THE REQUESTING TABLEE

To dispatch Nadel correctly, the system must know exactly which table requested service –reliably and in real time.

Service Request

A customer needs service — Call or Pay

Identify Table

Know which table created the request

Robot Task

Assign Nadel and navigate to the correct table

The critical gap: how do we reliably identify the table?

Design Criteria

1 Reliability

Works in crowded restaurants with people, chairs, plates, and changing lighting.

2 Real-time Performance

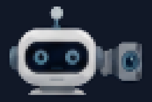
Should not slow down ROS2, SLAM, Nav2, sensors, UI, or voice interaction.

3 Clear Table Identity

The system must send an exact Table ID, not just guess from camera view.

EVALUATED APPROACHES: VISION-BASED TABLE REQUEST DETECTION

Before choosing the table module, we explored camera-based methods to identify which table requested service.



Approach 1: Robot Camera + ArUco Marker + Hand Raise

How it works

- Each table has an ArUco marker as a unique Table ID
- Customer raises hand to request service
- Robot detects gesture, then reads ArUco marker to identify the table

Why considered

- Simple and lightweight
- Easy to implement
- Gives a clear visual Table ID

Main Limitations

- Marker must be clearly visible to robot camera
- Blocked by customers, plates, cups, chairs, bad angles, or weak lighting
- Robot may not face the table while moving
- False positive feedback

Approach 2: Fixed Camera Monitoring

How it works

- Fixed camera placed in restaurant monitors tables from a wider view
- Detects raised hands or service requests from above

Why considered

- Wider view than the robot camera
- Can monitor multiple tables simultaneously

Main Limitations

- Blind spots and occlusion still occur
- Requires installation and calibration
- Raises privacy concerns
- Adds continuous video processing and network dependency

Key Takeaway: Both vision-based approaches were useful for testing, but not reliable enough as the main customer request system in a dynamic restaurant environment.

TABLE CALL / PAY SYSTEM

Dedicated ESP32 hardware modules placed on every table allow customers to instantly summon Nadel without needing staff intervention.

1. Customer Action

Patron presses physical "Call" or "Pay" button on the table module.

2. Signal Transmission

ESP32 transmits the unique Table ID and request type over local WiFi to the central server.

3. Robot Assignment

Server queues request. Nadel autonomously navigates to the specific Table ID to assist.

PROTOTYPE DEVELOPMENT JOURNEY

Concept
Ideation & Spec defining

CAD Design
Mechanical modeling

3D Printing
Fabrication of parts

Integration
Electronics assembly

Software Dev
UI and mapping

Testing
Real-world evaluation

PROTOTYPE COST

RPLIDAR A1M8 (360° LiDAR)	$177
Intel RealSense depth camera D435i	$490
Wheels (motors + wheel assemblies) + 2 motor drivers + 2 caster wheels	$220
1 x (10") touch screen (touch display)	$240
NVIDIA Jetson Orin Nano (dev kit / edge module)	$250
Model body (3-D printing)	$400
TOTAL	**$2000 - $2500**

New robot

🎯Target Market

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📈Revenue Streams

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- **Software Services:** Cloud analytics and premium AI features.

FUTURE WORK & UPGRADES

- **Autonomous Charging Dock:** Enabling the robot to detect low battery and automatically return to a charging station.
- **Computer Vision:** Implementing visual semantic SLAM to recognize specific tables visually rather than relying solely on 2D coordinates.
- **Fleet Management System:** Coordinating multiple Nadel robots on the same floor to prevent deadlocks and optimize delivery queues.
- **Analytics Dashboard:** Providing restaurant managers with insights on delivery times, peak hours, and robot utilization.
- **New design:** new user-friendly design making the user more comfortable interacting with nadel.

Thank You

We are ready for your questions and discussion.

Nadel – Fully Autonomous Service Waiter Robot

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